

Advanced (Habanero)

Q1. Solve the system: $x + y = 4$, $x - y = 2$

- (A) $x = 3, y = 1$
- (B) $x = 2, y = 2$
- (C) $x = 1, y = 3$
- (D) $x = 4, y = 0$
- (E) $x = 0, y = 4$

Q2. Solve the system: $2x + 3y = 7$, $x - 2y = -3$ using substitution

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) $x = -1, y = -2$

Q3. Solve the system: $x + 2y = 5$, $3x - 2y = 7$ using elimination

- (A) $x = 3, y = 1$
- (B) $x = 2, y = 2$
- (C) $x = 1, y = 3$
- (D) $x = 4, y = 0$
- (E) $x = 0, y = 4$

Q4. What is the solution to the system: $x + y = 4$, $x + y = 5$?

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q5. Solve the system: $x - 2y = 3$, $2x + 4y = 2$

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) No solution

Q6. Determine the number of solutions for the system: $x + y = 4$, $2x + 2y = 8$

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Q7. Solve the system: $2x - 3y = 5$, $x + 2y = 3$ using substitution

- (A) $x = 1, y = 2$
- (B) $x = 2, y = 1$
- (C) $x = 3, y = 0$
- (D) $x = 0, y = 3$
- (E) $x = -1, y = -2$

Q8. What is the solution to the system: $x - y = 2$, $x - y = 3$?

- (A) One solution
- (B) No solution
- (C) Infinitely many solutions
- (D) Two solutions
- (E) Three solutions

Open Ended Questions (FRQ)

Q9. Solve the system: $x + 2y - z = 3$, $2x - y + 2z = 5$, $3x + y - z = 2$ using elimination or substitution. Show all work.

Q10. A bakery is making a special batch of cookies for a holiday sale. They need to package 480 cookies into boxes that hold either 12 or 18 cookies. If they use x boxes that hold 12 cookies and y boxes that hold 18 cookies, the system of equations is: $12x + 18y = 480$, $x + y = 25$. Solve for x and y . Show all work.

Chef's Tips

- Tip 1: Make sure to check your work.
- Tip 2: Read each question carefully.
- Tip 3: Use the correct method to solve each system.